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🌐 TAriasVergara

Dr.-Ing. Tomás Arias-Vergara

Ph.D. in Computer Science

Personal information

Nationality: Colombian

Year of birth: 1991

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Google Scholar ID: rLGzEMsAAAAJ

Language proficiency:

Spanish: native ○ **English:** C1 ○ **German:** A2

Education

2022 – Ph.D. in Computer Science (Summa cum laude)

- Bi-national program: Universidad de Antioquia (UdeA), Colombia & Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany

2017 – Master degree in Engineering (with distinction)

- Universidad de Antioquia (Colombia)

2014 – Bachelor degree in Electronic Engineering

- Universidad de Antioquia (Colombia)

Current position

2023 - present – Postdoctoral researcher

- Pattern Recognition Lab
- Friedrich-Alexander-Universität Erlangen-Nürnberg (Germany)

Previous positions

2024 – Visiting scholar

- Topic: AI methods to analyze speech production of tongue cancer patients using cine-magnetic resonance imaging data
- Gordon Center for Medical Imaging
- Mass General Hospital & Harvard Medical School (USA)

2021 - 2023 – Research scientist

- Topic: Machine learning methods for the analysis of voice disorders using high-speed videoendoscopy data
- Department of Otorhinolaryngology, Head and Neck Surgery
- University Hospital Erlangen (Germany)

2018 - 2021 – Marie Curie Research Fellow

- Topic: Automatic methods for the analysis of speech production of people with cochlear implants
- Department of Otorhinolaryngology, Head and Neck Surgery
- Ludwig-Maximilians Universität München (Germany)



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Teaching

2024 – Present

- Graduate: *Deep Learning, Seminar Automatic Analysis of Voice, Speech and Language Disorders in Speech Pathologies, Project Time Series*
- FAU (Germany)

2022 – 2023

- Undergraduate: *Signal Processing I*
- Graduate: *Pattern Analysis*
- UdeA (Colombia)

2017 – 2019

- Undergraduate: *Digital Signal Processing (On site/Virtual Program) & Signal Processing III (Virtual Program)*
- UdeA (Colombia)

Academic activities

Activities as reviewer

Journals (selected)

IEEE Journal of Biomedical and Health Informatics ◦ Biomedical Signal Processing and Control ◦ Expert Systems with Applications ◦ Computer Methods in Biomechanics and Biomedical Engineering ◦ Movement disorders ◦ Speech Communication ◦ Sensors ◦ Scientific Reports ◦ IEEE Access ◦ Diagnostics ◦ Journal of Speech, Language, and Hearing Research (JSLHR) ◦ Journal of Voice ◦ Pattern Recognition Letters ◦ Nature: Scientific Data ◦ Computer Speech & Language ◦ Digital Signal Processing

Conferences

International Conference on Acoustics, Speech, & Signal Processing (ICASSP 2023, 2024, 2025, 2026) ◦ IEEE Automatic Speech Recognition and Understanding Workshop (ASRU 2022) ◦ Workshop on Engineering Applications (WEA 2019) ◦ Text, Speech & Dialogue (TSD 2023, 2025, 2026) ◦ Workshop on Multimodal Healthcare Data (International Conference on Machine Learning - ICML, 2023) ◦ Interspeech (2023, 2024, 2025, 2026) ◦ International Symposium on Biomedical Imaging (ISBI 2024, 2025)

Organization of academic events

2025 - Local Organizing Committee Co-chair. 28th International Conference on Text, Speech, and Dialogue (TSD). Erlangen, Germany.

2024 - Organizing Committee Co-chair. 5th International Workshop on Multiscale Multimodal Medical Imaging and 1st Workshop on Machine Learning for Multimodal/-sensor Healthcare Data. Co-located with MICCAI 2024

2023 - Technical Assistant. 1st Workshop on Machine Learning for Multimodal Healthcare Data. Co-located with ICML 2023



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Projects/Funding

2025 - 2027 – DFG

- Role: Principal Investigator
- Title: The effect of social interactions on (dis)honest communication.
- Budget: 233.264 EUR
- Funding agency: Deutsche Forschungsgemeinschaft (DFG)

2024 - 2025 – Emerging Talents Initiative (FAU)

- Role: Principal Investigator
- Title: A multimodal approach for automatic generation of radiology reports using chest X-ray images, clinical free-text, and spoken commands.
- Budget: 14.744 EUR
- Funding agency: Friedrich-Alexander Universität Erlangen-Nürnberg

2024 - 2027 – DFG

- Role: Project Member
- Title: Coordinated network protection based on machine learning methods.
- Funding agency: Deutsche Forschungsgemeinschaft (DFG)

Supervision

Completed the supervision of 38 Master's/Bachelor's theses and currently I'm supervising 10 Master's/Bachelor's students and three doctoral students (scientific supervision). In addition, I'm leading the Time Series Intelligence Colloquium at Pattern Recognition Lab (FAU, Erlangen).

Other academic activities

- Oral session chair at Interspeech 2025 (Rotterdam, Netherlands)
- Oral session chair at Interspeech 2024 (Kos, Greece)
- Oral session chair at ICVPB 2024 (Erlangen, Germany)
- Oral session chair at Interspeech 2023 (Dublin, Ireland)
- Poster session chair at Interspeech 2022 (Incheon, South Korea)